DHARMSINH DESAI UNIVERSITY
FACULTY OF TECHNOLOGY, NADIAD
COMPUTER ENGINEERING DEPARTMENT
B.TECH. CE SEM-IV
SUBJECT: JAVA TECHNOLOGY

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Lab 2

Aim: To study and use String, StringBuffer, StringBuilder, array of objects, this keyword, constructor overloading.

Programming Exercise

1) **Write a program that returns the number of times the string "hi" appears anywhere in the given string.**

ANS.

```
import java.util.Scanner;

public class CountHi {
	public static void main(String[] args) {
		Scanner sc = new Scanner(System.in);

		System.out.print("Enter a String: ");
		String str = sc.next();

		int count = 0;

		for(int i=0; i<str.length()-1; i++) {
			if(str.substring(i,i+2).equals("hi")) {
				count++;
			}
		}

		System.out.println("The no. of 'hi' appears in given String is:" + count);
```

```
        }
}
```

Output:

```
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ gedit CountHi.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ javac CountHi.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java CountHi
Enter a String: hivimalhixyzhihi
The no. of 'hi' appears in given String is:4
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$
```

2) **Write a program which check whether the input string is palindrome or not and display an appropriate message [e.g. "Refer" is a Palindrome].**

ANS.

```
import java.util.Scanner;

public class PalindromeCheck {
        public static void main(String[] args) {
                Scanner sc = new Scanner(System.in);

                System.out.print("Enter a String: ");
                String str = sc.next();

                str = str.toLowerCase();

                int left = 0, right = str.length() - 1;
                boolean isPalindrome = true;

                while(left < right) {
                        if( str.charAt(left) != str.charAt(right)) {
                                isPalindrome = false;
                                break;
                        }
                        left++;
                        right--;
                }

                if(isPalindrome) {
                        System.out.println(str + " is a palindrome string.");
                } else {
                        System.out.println(str + " is NOT a palindrome string.");
                }
```

```
        }
}
```

Output:

```
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ gedit PalindromeCheck.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ javac PalindromeCheck.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java PalindromeCheck
Enter a String: refer
refer is a palindrome string.
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java PalindromeCheck
Enter a String: vimal
vimal is NOT a palindrome string.
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$
```

3) **Write a program that takes your full name as input and displays the abbreviations of the first and middle names except the last name which is displayed as it is. For example, if your name is Robert Breet Roser, then the output should be R.B.Roser.**

ANS.

```
import java.util.Scanner;

public class NameAbbreviation {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);
    System.out.print("Enter your full name: ");
    String fullName = sc.nextLine();

    String[] parts = fullName.split("\s+");
    int n = parts.length;
    StringBuilder sb = new StringBuilder();
    for (int i = 0; i < n - 1; i++) {
      sb.append(Character.toUpperCase(parts[i].charAt(0))).append(".");
    }

    sb.append(parts[n - 1]);

    System.out.println("Abbreviated form: " + sb.toString());
  }
}
```

Output:

```
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ gedit NameAbbreviation.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ javac NameAbbreviation.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java NameAbbreviation
Enter your full name: Vimal Dilipbhai Solanki
Abbreviated form: V.D.Solanki
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$
```

4) **Write a method String removeWhiteSpaces(String str) method that removes all the white spaces from the string passed to the method and returns the modified string. Test the functionalities using the main() method of the Tester class.**

ANS.

```
import java.util.Scanner;

public class Tester {

	public static String removeWhiteSpaces(String str){
		return str.replaceAll("\s+","");
	}

	public static void main(String[] args){
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter a String : ");
		String str = sc.nextLine();

		String result = removeWhiteSpaces(str);
		System.out.println("Original String: "+ str);
		System.out.println("Replaced Sting is: " + result);
	}
}
```

Output:

```
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ gedit Tester.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ javac Tester.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java Tester
Enter a String : Java Programming Practice Lab
Original String: Java Programming Practice Lab
Replaced Sting is: JavaProgrammingPracticeLab
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$
```

5) **Write a class Student with member int roll_no, String name, and an array to store marks of 5 subjects. Demonstrate constructor overloading and use this keyword. Write a findAverage() method that return double value. Write a TestStudent class containing main() method to do the following:**
   a. **Store the details of one student by creating one object of the class student and display them.**
   b. **Store the details of 3 student by creating an array of Objects of student class and display the details of student who has the highest average amongst the three students.**

ANS.

<u>Student.java</u>

```
public class Student {
  private int roll_no;
  private String name;
  private int[] marks;

  public Student() {
     this.roll_no = 0;
     this.name = "Unknown";
     this.marks = new int[5];
  }
  public Student(int roll_no, String name, int[] marks) {
     this.roll_no = roll_no;
     this.name = name;
     this.marks = marks;
  }
  public double findAverage() {
     int sum = 0;
     for (int m : marks) {
        sum += m;
     }
     return sum / 5.0;
  }
  public void display() {
     System.out.println("Roll No: " + roll_no);
     System.out.println("Name   : " + name);
     System.out.println("Average: " + findAverage());
  }
}
```

TestStudent.java

```
import java.util.Scanner;

public class TestStudent {
	public static void main(String[] args) {
		Scanner sc = new Scanner(System.in);

		// part A
		System.out.println("Enter details for student:");
		System.out.print("Roll Number: ");
		int roll = sc.nextInt();
		System.out.print("Name: ");
		sc.nextLine();
		String name = sc.nextLine();
		int[] marks = new int[5];
		System.out.print("Enter 5 subject marks separated by space: ");
		for (int i = 0; i < 5; i++) {
			marks[i] = sc.nextInt();
		}
		Student s1 = new Student(roll, name, marks);
		System.out.println("\nDetails of Student:");
		s1.display();

		// part B
		Student[] stu = new Student[3];
		for(int i=0; i<3; i++) {
			System.out.println("\nEnter details for student " + (i + 1) + ":");
			System.out.print("Roll Number: ");
			roll = sc.nextInt();
			System.out.print("Name: ");
			sc.nextLine();
			name = sc.nextLine();
			marks = new int[5];
			System.out.print("Enter marks of 5 Subject :");
			for (int j = 0; j < 5; j++) {
			marks[j] = sc.nextInt();
			}
			stu[i] = new Student(roll, name, marks);

		}
		double maxAvg = 0;
		int maxindex = 0;
		for (int i = 0; i < 3; i++) {
		  double avg = stu[i].findAverage();
		  if (avg > maxAvg) {
		     maxAvg = avg;
		     maxindex = i;
		  }
```

```
            }
            System.out.println("\nStudent with Highest Average:");
            stu[maxindex].display();
        }
}
```

Output:

```
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ gedit TestStudent.java
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$ java TestStudent
Enter details for student:
Roll Number: 1
Name: Vimal Solanki
Enter 5 subject marks separated by space: 98 98 98 98 98

Details of Student:
Roll No: 1
Name   : Vimal Solanki
Average: 98.0

Enter details for student 1:
Roll Number: 10
Name: dhyey
Enter marks of 5 Subject :97 98 96 90 95

Enter details for student 2:
Roll Number: 11
Name: hanuj
Enter marks of 5 Subject :96 95 94 96 94

Enter details for student 3:
Roll Number: 13
Name: Darsh
Enter marks of 5 Subject :78 89 80 90 98

Student with Highest Average:
Roll No: 10
Name   : dhyey
Average: 95.2
user1@celab8-ThinkCentre-M720t:~/Desktop/CE4_158$
```